# Internet of Things Laboratory - Experiments and Examples to Import

V Semester | Course Code: EC2601-1 | 2026-27

| Sl. No. | Experiment Name | Board | Example(s) to Import |
|---|---|---|---|
| 1 | Peripheral Interfacing using GPIO, LCD and USB HID | NXP FRDM-RW612 | **frdmrw612_dev_hid_generic_bm** (USB HID generic, bare-metal SDK example) |
| 2 | Implementation of Proprietary Wireless Networks Using the Silicon Labs Connect Stack | 2 × BRD2705A (EFR32xG28 Explorer Kit) | **connect_soc_sensor** (End Node)<br>**connect_soc_sink** (Coordinator) |
| 3 | Integrated Thread Protocol Analysis with CLI Configuration and Sleepy End Device Power Management | MGM260P-EK2713A (BRD2713A) | **sleepy-demo-ftd** (Full Thread Device)<br>**sleepy-demo-mtd** (Sleepy End Device) |
| 4 | Wi-Fi Connectivity and an Embedded HTTP Web Server on the NXP FRDM-RW612 | NXP FRDM-RW612 | **wifi_examples/wifi_httpsrv** (lwIP + FreeRTOS HTTP server) |
| 5 | Implementation and Evaluation of Wi-SUN FAN Topologies Using Border Routers | 2 × BRD2705A (EFR32xG28 Explorer Kit) | **wisun_soc_brcli** (Wi-SUN SoC Border Router CLI)<br>**wisun_soc_ping** (Wi-SUN SoC Ping node) |
| 6 | MQTT and HTTP Communication over Wi-Fi on the NXP FRDM-RW612 | NXP FRDM-RW612 | **wifi_examples/wifi_mqtt** (lwIP MQTT client + FreeRTOS)<br>(HTTP part reuses **wifi_httpsrv** from Experiment 4) |
| 7 | Bluetooth Low Energy Communication using NXP RW612 | NXP FRDM-RW612 | **frdmrw612_wireless_uart** (EdgeFast Bluetooth Wireless UART) |
| 8 | Implementation of Matter over Wi-Fi/Thread | NXP FRDM-RW612 | **frdm_rw612_light_switch_otbr_combo** (Application Code Hub project; contains **light-switch-combo-app**, Matter + OpenThread Border Router) |

### Note: Bootloader for the Silicon Labs boards

| Experiments | Board | Bootloader to import (Simplicity Studio) |
|---|---|---|
| 2 and 5 | BRD2705A (EFR32xG28, 1024 KB flash) | **Bootloader - SoC Internal Storage (single image on 1MB device)** |
| 3 | MGM260P-EK2713A (EFR32MG26, ~3.2 MB flash) | **Bootloader - SoC Internal Storage (single image on 3200kB device)** |

**Why it is required:** The Silicon Labs example projects (Experiments 2, 3 and 5) are built to start *after* a bootloader, which sits at the beginning of flash and hands control to the application. If the board has no bootloader (for example after a full device erase), the application has nothing valid to start from and the board may not run even though flashing succeeds. The bootloader must also match the board's flash size, because its storage slot is sized and placed for that device; a mismatched one (such as the 4MB-flash sample) will not fit the board.

**Order:** Create the bootloader project for the board (select the board under Compatible Device), build it and flash it first. Then flash the application example. This is needed once per board unless the device is erased. Many Explorer Kits ship with a bootloader already present; if the example runs normally after flashing, no extra bootloader step is needed.

Example names are taken from the Procedure and Development Board sections of the IoT Laboratory Report.